

Test Plan

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December 9, 2025

Version 1.0



Status

Reviewed	David Nordlund	2025-11-14
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DOCUMENT HISTORY

Version	Date	Changes made	Sign	Reviewer
0.1	2025-11-14	Initial test plan drafted from approved requirements	AF, AG, VG, MM	DN
0.2	2025-12-16	Updated after implementation of core pipeline and GUI prototype	AF, AG, VG, MM	DN
1.0	2025-12-19	Final version after execution of module and system tests	AF, AG, VG, MM	DD

1 INTRODUCTION

The purpose of this document is to describe the **test plan** for the *Automatic Music Transcription (AMT) System*, developed within the TSKS23 CDIO project.

The AMT system converts piano performances captured through *microphone input* (Casio CTK-3500 keyboard + MIC ARM) or *pre-recorded audio files* into a structured list of musical notes (e.g. C4, D#5, A5). The core processing pipeline performs

- onset and offset detection,
- pitch estimation (monophonic and limited polyphonic),
- note duration computation,
- storage of transcription results in an SQLite database,
- and presentation of results via a PyQt5-based GUI.

This test plan is based on the approved *Requirements Specification* (Tollgate 1) and the current *Design Specification*. It defines how we will verify that the implemented system fulfils the functional and non-functional requirements, with a focus on:

- correctness of the detection algorithms (Module 1: input handling and core processing engine),
- usability and correctness of the graphical user interface (Module 2: GUI),
- end-to-end behaviour from audio input to stored and exported transcription results.

The tests are planned and executed in the LiU lab environment (room 3D:502) and, when relevant, in more realistic acoustic environments (e.g. corridors or outdoors) to evaluate robustness. Both synthetic reference signals (generated from MIDI) and real recordings from the Casio keyboard are used as test material.

2 TEST METHODOLOGY

The testing strategy follows a bottom-up approach aligned with the project phases and activity plan:

1. **Module tests** of the core detection algorithms (onset/offset, pitch, duration) and input handling (file loading and recording).
2. **Module tests** of the graphical user interface (GUI) for file selection, recording, visualisation, and export.
3. **Integration tests** where Module 1 and Module 2 are executed together, including database storage and history retrieval.
4. **System-level tests** that evaluate end-to-end behaviour against key performance requirements such as pitch accuracy, timing accuracy, and processing time.

Test design is directly traceable to the requirements defined in the Requirements Specification (e.g. pitch accuracy, supported input formats, latency, robustness to noise). For each test we specify:

- a unique **Test ID**,
- **Test Description** and relation to one or more requirements,
- **Location** (lab room or environment),
- **Target Week** in the project schedule,
- and **Status** (Planned, In progress, Completed).

Table 1 summarises the main planned tests, including both module and system-level activities.

Test ID	Test Description	Location	Week	Status
IO-01	Verify audio recording and file loading for WAV inputs (correct sample rate, duration, and error handling)	Lab 3D:502	46	Planned
IO-02	Verify reference MIDI loading and mapping to internal note representation	Lab 3D:502	46	Planned
ONSET-01	Onset detection accuracy on single-note sequences (mean error ≤ 50 ms)	Lab 3D:502	47	Planned
OFFSET-01	Offset detection accuracy on sustained and short notes (mean error ≤ 70 ms)	Lab 3D:502	47	Planned
PITCH-01	Pitch detection for single tones across the keyboard range (accuracy $\geq 80\%$)	Lab 3D:502	48	Planned
POLY-01	Two-note chord transcription accuracy (accuracy $\geq 60\%$)	Lab 3D:502	48	Planned
DUR-01	Note duration estimation using MIDI as reference (MAE ≤ 0.1 s)	Lab 3D:502	49	Planned
PERF-01	Runtime test for 30 s audio clip (processing time ≤ 10 s on lab PC)	Lab 3D:502	50	Planned
GUI-01	Validate GUI workflow: record, process, visualise, and export transcription	Lab 3D:502	50	Planned
INT-01	End-to-end transcription including database storage and history retrieval	Lab 3D:502	51	Planned

Table 1: Test plan for Automatic Music Transcription System validation

3 TESTS FOR MODULE 1 – CORE DETECTION ALGORITHMS

Module 1 covers the input handling and core detection algorithms of the AMT system: loading/recording audio, preprocessing, onset/offset detection, pitch estimation (YIN/pYIN), and duration computation. The tests below are designed to verify that these algorithms meet the quantitative requirements from the Requirements Specification.

4 TESTS FOR MODULE 2 – GRAPHICAL USER INTERFACE

Module 2 comprises the graphical user interface (GUI) implemented in PyQt5. It connects the user to the underlying transcription engine and database, providing controls for recording, file loading, processing, visualisation, and export. The following tests focus on interaction flows and user-visible behaviour rather than algorithmic details.

5 TESTS FOR MODULE 3 – BACKEND PROCESSING ENGINE

Module 3 comprises the core backend processing engine responsible for executing the transcription pipeline: pre-processing, onset/offset detection, pitch estimation (YIN/pYIN), multi-pitch extraction, note grouping, and post-processing. These tests verify the correctness, stability, and performance of the pipeline.

6 TESTS FOR MODULE 4 – DATABASE MANAGEMENT SYSTEM

Module 4 provides persistent storage for audio metadata, transcriptions, notes, and processing logs using SQLite. Tests ensure data integrity, correct schema behaviour, and robustness of insert/update/delete operations during active transcription sessions.

Test ID	Test Objective	Req. #	Location	Week	Priority
M1-PITCH-01	Verify pure-tone pitch detection and correct note mapping (e.g. C4, A4, G5) over the supported keyboard range	2, 6	Lab 3D:502	46	P1
M1-HARM-01	Validate that harmonic overtones are not misclassified as separate notes when using YIN/pYIN	1, 2	Lab 3D:502	47	P1
M1-ONSET-01	Measure onset detection accuracy (mean error ≤ 50 ms) on short melodies using MIDI-generated reference audio	2.3.4	Lab 3D:502	47	P1
M1-OFFSET-01	Evaluate offset detection accuracy (mean error ≤ 70 ms) for short and sustained notes	2.3.5	Lab 3D:502	48	P1
M1-DUR-01	Validate note duration estimation (MAE ≤ 0.1 s) by comparing to reference MIDI note lengths	2.3.6	Lab 3D:502	49	P2
M1-ROBUST-01	Assess robustness of detection under moderate background noise (different SNR levels, e.g. corridor, hallway)	15	Lab / Outdoors	50	P3

Table 2: Module 1 – Core detection algorithms test plan

7 TESTS FOR MODULE 5 – OUTPUT AND EXPORT SYSTEM

Module 5 handles final user-visible output: transcription formatting, visualisation (waveform + spectrogram), text/MIDI export, and review tools. These tests ensure visualisation correctness, export accuracy, and consistency with back-end results.

Test ID	Test Description	Success Criteria	Location	Week
M2-GUI-01	Verify live recording controls (start, stop, pause) during audio capture	All control buttons respond within 0.5 s and no crashes occur	Lab 3D:502	46
M2-GUI-02	Test file upload for supported formats (.wav, .mid)	Selected files load successfully and basic metadata (duration, sample rate) is shown	Lab 3D:502	47
M2-GUI-03	Validate real-time visualisation of detected notes during playback/recording	Detected notes and waveforms update within 1 s of the audio event	Lab 3D:502	48
M2-GUI-04	Check parameter adjustment (thresholds, sensitivity) and dynamic update in processing results	Changing parameters leads to visible changes in onset/pitch detection without restarting the GUI	Lab 3D:502	49
M2-GUI-05	Verify waveform and spectrogram visualisation	Both plots are clear, correctly scaled, and synchronised with the audio timeline	Lab 3D:502	50
M2-GUI-06	Test export of transcription results (text/MIDI) and integration with database history	Exported files open correctly and correspond to the notes stored in the database	Lab 3D:502	51

Table 3: Module 2 – Graphical user interface test plan

Test ID	Test Objective	Req.#	Location	Week	Priority
M3-PRE-01	Verify preprocessing pipeline (normalization + HPF) produces identical output for repeated runs (determinism test)	2.1	Lab 3D:502	46	P1
M3-PIPE-01	Validate full transcription pipeline (onset → offset → pitch → duration) using BandLab synthetic audio	2, 3, 6	Lab 3D:502	47	P1
M3-YIN-01	Test YIN stability on long sustained notes (14s in Piano_Bomb_G#2) – no drift > 5 Hz over duration	6.1	Lab 3D:502	47	P1
M3-YIN-02	Evaluate YIN threshold tuning (frame rejection) for low-energy segments (fade-out tails)	6.2	Lab 3D:502	48	P2
M3-POLY-01	Validate two-note polyphonic detection using filtered BandLab loops (accuracy 60%)	7	Lab 3D:502	48	P1
M3-GROUP-01	Verify onset grouping merges adjacent events correctly (gap < 50 ms → single note)	2.3.4	Lab 3D:502	49	P2
M3-DUR-01	Check duration post-processing (MAE 0.1 s) using MIDI alignment from load_midi_file()	2.3.6	Lab 3D:502	49	P2
M3-PERF-01	Runtime test: full transcription of 20 s audio must finish in 8 s (workstation benchmark)	12	Lab 3D:502	50	P1
M3-ROBUST-01	Stress test with clipped audio + background noise to verify system recovery without crashing	15	Lab / Outdoors	50	P3

Table 4: Module 3 – Backend processing engine test plan

Test ID	Test Objective	Req.#	Location	Week	Priority
M4-DB-01	Verify database creation (all tables created exactly as schema) on first program startup	10.1	Lab 3D:502	47	P1
M4-DB-02	Validate insertion of audio metadata (filename, duration, sample rate, timestamp)	10.2	Lab 3D:502	48	P1
M4-DB-03	Verify transcription events stored with correct foreign keys (audio_file_id references audio_files.id)	10.3	Lab 3D:502	48	P1
M4-DB-04	Test note-level insertion into transcription_notes (pitch, note name, index in chord)	10.3	Lab 3D:502	49	P2
M4-DB-05	Validate cascading deletion: deleting an audio file removes all transcription + note rows	10.4	Lab 3D:502	49	P2
M4-DB-06	Stress test for 50+ consecutive transcription sessions (no corruption, no locking errors)	12	Lab 3D:502	50	P2
M4-DB-07	Validate search + filter queries in History Tab (latency 200 ms)	11	Lab 3D:502	50	P3
M4-DB-08	Verify logging of warnings/errors into processing_logs when audio load fails, or pitch detection crashes	14	Lab 3D:502	51	P1

Table 5: Module 4 – Database management system test plan

Test ID	Test Objective	Req.#	Location	Week	Priority
M5-VIS-01	Verify waveform plot matches actual audio amplitude (no scaling or offset errors)	8.1	Lab 3D:502	48	P2
M5-VIS-02	Validate spectrogram rendering (STFT correctness, logarithmic scale, colour mapping)	8.1	Lab 3D:502	48	P2
M5-OUT-01	Validate human-readable transcription output format (onset, offset, duration, name, pitch Hz)	9	Lab 3D:502	49	P1
M5-OUT-02	Verify export of transcription to TXT (correct formatting, line count = number of events)	9	Lab 3D:502	49	P1
M5-OUT-03	Verify export to MIDI (pitch numbers must match note mapping from Module 1/3)	9	Lab 3D:502	50	P1
M5-OUT-04	Compare GUI-displayed transcription vs. exported file (must match 1:1)	8, 9	Lab 3D:502	50	P2
M5-OUT-05	Validate history export (batch export) correctness for 10 files	11	Lab 3D:502	51	P3

Table 6: Module 5 – Output and export subsystem test plan